Site Speed Optimization

Q-1. Use Google PageSpeed Insights to analyze the homepage of your favorite food delivery app (like Zomato or Swiggy) and note down its LCP, INP, and CLS scores.

Ans:

Metric	Meaning	Desktop	Mobile
LCP	Largest Contentful Paint	1 second	2.3 seconds
INP	Interaction to Next Paint	62 ms	158 ms
CLS	Cumulative Layout Shift	0	0

Explanation:

- **LCP:** Measures how quickly the largest visible content loads.
- **INP:** Measures how quickly a webpage responds to user interactions.
- **CLS:** Measures unexpected movement of page elements during loading.

Note: These are the sample scores given in the assignment. Verify the current scores using Google PageSpeed Insights before submitting.

Q-2. Compress an image (JPG or PNG) you use on a personal website or blog using TinyPNG or Squoosh, then compare the original and optimized file sizes and loading times.

Ans:

Detail	Original Image	Optimized Image
File format	JPG	JPG
File size	42.2 KB	21.2 KB

Loading time	0.80 seconds	0.48 seconds
Reduction	—	56%

Explanation:

Image compression reduces file size while maintaining acceptable image quality. Smaller images can load faster and improve website performance.

Note: These are sample results. Compress your own image using TinyPNG or Squoosh and check its actual file size and loading time.

Q-3. Add a simple caching rule to your website by editing the .htaccess file to set browser cache for images to 1 month.

Ans:

Add the following code to your `.htaccess` file:

```
<IfModule mod_expires.c>
  ExpiresActive On

  # Cache JPEG images for 1 month
  ExpiresByType image/jpeg "access plus 1 month"

  # Cache PNG images for 1 month
  ExpiresByType image/png "access plus 1 month"
</IfModule>
```

Explanation:

- `ExpiresActive On` enables expiration rules.
- `ExpiresByType image/jpeg` sets browser caching for JPEG images.
- `ExpiresByType image/png` sets browser caching for PNG images.
- `access plus 1 month` allows the browser to cache these images for one month.

This can reduce repeat downloads and improve loading speed for returning visitors.

Q-4. Sign up for a free Cloudflare account and connect it to a test website or a personal portfolio. Enable CDN and

caching, then check if your site loads faster using any online speed test tool.

Ans:

Test	Before Cloudflare	After Cloudflare
Page Load Time	3.2 seconds	2.1 seconds
Performance Score	68	82
CDN	No	Yes
Browser Caching	Basic	Enabled

Explanation:

Cloudflare's CDN delivers cached website content through its global network. Browser caching stores certain resources locally, reducing repeat downloads. These features can improve website loading speed.

Note: These are sample results from the assignment. Connect your own test website to Cloudflare and record the actual results from a speed-testing tool.

Q-5. Create a table listing what LCP, INP, and CLS mean, and for each, give one real example of a bad user experience from an app you use (like Flipkart, Instagram, or BookMyShow) when that metric is poor.

Ans:

Metric	Full Form	What It Measures	Example of Poor User Experience
LCP	Largest Contentful Paint	How quickly the largest visible content loads	On Flipkart, a large product banner takes several seconds to appear, leaving the user waiting.
INP	Interaction to Next Paint	How quickly a page responds to user interactions	On Instagram, a user taps the Like button, but the interface takes noticeable time to respond.

CLS	Cumulative Layout Shift	How much visible content unexpectedly moves during loading	On BookMyShow, an image or advertisement loads and shifts a movie option, causing the user to click the wrong element.

Conclusion:

Good LCP, INP, and CLS performance helps websites load quickly, respond smoothly, and maintain a stable layout, improving the overall user experience.

Add a proper assignment headingReplace sample results with placeholders